

Generation Interconnection System Impact Study Report

For Queue Project Q20-1620

Fuller Road 138 kV

October 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1620. The Interconnection Customer proposes a 109.6 MW solar photovoltaic facility interconnecting at the Fuller Road 138 kV substation in the Virginia area. The request was received on December 09, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 411 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$776,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bateville - Innovation 138 KV	Medical - Eagleville 138 KV	116.2	279	32.4
Bateville - Novec Country Club 138 KV	Medical - Eagleville 138 KV Line	108.8	236	32.4
8Th Street Cable Terminal - Newdale 138 KV	Newdale - 8Th Street Cable Terminal 138 KV #1	105.8	341	23.1
8Th St Cable Terminal - Carreon 138 KV Line 1	8Th Steet Cable Terminal - Carreon 138 KV #2	103.2	314	22.4
Newdale - 8Th St Cabl Terminal 138 KV	8Th St Cable Terminal - Newdale 138 KV Cir. 2	100.0	361	23.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Innovation - Bateville 138 KV Line	Rebuild line with higher-capacity conductor	\$1,315,000	11% (\$149,000)
Bateville - Novec Country Club 138 KV	Reconductor line	\$1,736,000	11% (\$197,000)
8Th Street Cable Terminal - Newdale 138 KV	Reconductor line	\$688,000	15% (\$105,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
8Th Street Cable Terminal - Carreon 138 KV	Reconductor line	\$1,289,000	15% (\$196,000)
8Th Street Cable Terminal - Newdale 138 KV	Reconductor line	\$845,000	15% (\$129,000)
	Total allocated to project		\$776,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.